Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [12ME831]

EIGHTH SEMESTER B.E. EXAMINATION, JUNE - 2016

NON CONVENTIONAL ENERGY SYSTEMS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	What is meant by non-conventional energy sources? Explain in brief these energy sources with special reference to Indian context.	07
	b.	What are the advantages and limitations of renewable energy sources?	07
2.	a.	Explain the working principle of two basic types of instruments employed for the measurement of solar radiation.	07
	b.	Determine the average value of solar radiation on a horizontal surface for June22, at the latitude of 10°N, if constants 'a' and 'b' are given as equal to 0.30 and 0.51 respectively and the ratio $\frac{\bar{n}}{N} = 0.55$	07
3.	a.	With a neat sketch, explain the components of a flat-plate solar collector.	07
	b.	Describe the working of solar furnace with a neat sketch. What are the advantages and limitations of a solar furnace?	07
4.	a.	Explain the logic principles of wind energy conversion.	07
	b.	Explain the classification of wind energy conversion systems.	07
5.	a.	Discuss the turbulence and acceleration effect in the wind plant.	07
	b.	Describe with a neat sketch the working of a cup type anemometer.	07
6.	a.	Briefly explain the biomass conversion technologies.	07
	b.	Describe the classification of biogas plants.	07
7.	a.	Describe the various methods of obtaining energy from biomass.	07
	b.	The following data are given for a family biogas digester suitable for the output of five cows: the retention time is 20 days, temperature 30°C, dry matter consumed /day =2kg, biogas yield =0.24m^3/kg. burner efficiency = 60%, methane proportion=0.8, heat of methane combustion =28MJ/m^3. Determine: i. vol. of digester ii. power available	07
8.	a.	What are the main types of OTEC power plants? Describe briefly their working.	07
	b.	List out the advantages and disadvantages of tidal power generation.	07